Counting involutions that generate a chiral polyhedron

Human Compression project

We give one counting argument for fourteen sporadic cases of the existing existence theorem of Leemans and Liebeck [1]. It uses ordinary class sizes and a complete list of maximal subgroup classes. Both are retained classification inputs, with their exact numerical use recorded. The same involution bound handles all fourteen cases without distinguishing the conjugacy classes of reversing automorphisms.

1 The common counting theorem

For a finite group H, let $i_d(H)$ be the number of elements of exact order d. Let G be nonabelian simple and identify G with its inner automorphism group in $A = \mathrm{Aut}(G)$. Fix $x \in G$ of odd prime order p. Choose one representative of each G-conjugacy class of maximal subgroups, and define

$$B_G(x) = |C_G(x)| \sum_M \frac{i_p(M) i_2(M)}{|M|}. \tag{1}$$

Let μ be an upper bound for $i_2(C_G(\alpha))$ for every involution $\alpha \in A$ that inverts x. Put

$$R_A(x) = \begin{cases} 0, & x \text{ is not conjugate to } x^{-1} \text{ in } A, \\ |C_A(x)|\mu, & \text{otherwise.} \end{cases} \tag{2}$$

The upper bound in (1) may be rational; it need not count an actual set exactly.

Theorem 1. *If $i_2(G) > B_G(x) + R_A(x)$, there is an involution $t \in G$ such that $G = \langle x, t \rangle$ and no automorphism of G inverts x while fixing t.*

Proof. Every maximal subgroup $M < G$ is self-normalizing. Indeed its normalizer is M or G, and the latter would make M a proper nontrivial normal subgroup of the simple group G. Count the conjugates of M containing x. The set of conjugating elements taking x into M has size $|C_G(x)|\,|x^G \cap M|$. Each conjugate of M is counted $|M|$ times. Hence the number of these conjugates is

$$\frac{|C_G(x)|\,|x^G \cap M|}{|M|} \leq \frac{|C_G(x)| i_p(M)}{|M|}.$$

If $\langle x, t \rangle$ is proper, it is contained in one of these maximal subgroups. Summing their involution counts proves that at most $B_G(x)$ involutions fail to generate with x.

If an automorphism α inverts x and fixes a generating partner t, then α^2 fixes both generators and is the identity. Since p is odd, α is nonidentity and thus an involution. All elements of A taking x to x^{-1}, if any exist, form a coset of $C_A(x)$: the quotient of any two such elements centralizes x. There are at most $|C_A(x)|$ possible inverting involutions. Each fixes at most μ involutions of G. The union of these fixed sets has size at most $R_A(x)$; it is empty if x is not conjugate to its inverse in A. The strict inequality therefore leaves an involution outside both excluded sets. It has the required properties. □

The maximal-subgroup count and inverter coset are standard counting arguments, closely related to the preliminary lemmas of [1]. Our common specialization uses a deliberately coarse bound for μ. It avoids a separate analysis of the reversing involutions in each group.

Lemma 2. *If $A = G$, one may take*

$$\mu = \max_{M \text{ maximal in } G} i_2(M). \tag{3}$$

Suppose instead that $[A : G] = 2$ and $|C_A(a)| = 2|C_G(a)|$ for every involution $a \in G$. Then one may take

$$\mu = \max_{M \text{ maximal in } A,\ M \neq G} i_2(M). \tag{4}$$

Proof. The centre of A is trivial. An automorphism central in A commutes with every inner automorphism; therefore it fixes every element of the centreless group G and is the identity. In particular the centralizer of a nonidentity involution in A is proper and lies in a maximal subgroup. This proves (3), since $C_G(\alpha) \leq C_A(\alpha)$.

For (4), if $\alpha \notin G$, its centralizer contains α and is not contained in G. If $\alpha \in G$, the stated centralizer index likewise shows that $C_A(\alpha)$ contains an element outside G. A maximal subgroup containing this centralizer therefore cannot be G. Containment bounds its involution count by (4), and again $C_G(\alpha) \leq C_A(\alpha)$. □

The centralizer condition in the second part is checked for every involution class of G, using its actual fusion into A. It is not inferred merely from the fact that A/G has order two.

2 Fourteen applications

For each group below choose the specified prime-order class in the bound data. The column L is a lower bound for $i_2(G)$, and B, R are the exact upper bounds computed from (1)–(4). In every row $B + R < L$. Only for B and M do we replace the exact involution count by a smaller power of ten in this display; the full counts remain in the certificate.

G	p	L	B	R
J_4	29	51747149311	29	101486051
Fi_{23}	23	12895196031	976120	0
Fi'_{24}	29	7824165773823	58	747921369827
M	71	10^{20}	4970	0
B	47	10^{12}	0	0
Th	31	976841775	44826	0
Ly	37	1296826875	74	15054375
Co_1	23	12792894975	72100538	0
Co_2	23	29771775	7590	0
Co_3	23	2778975	7590	0
Ru	29	1846575	870	390195
Fi_{22}	13	37706175	1443546	18669235
$\mathrm{O'N}$	31	2857239	1860	95945
HN	19	75603375	7182	2662869

The full automorphism groups equal G except in the four rows $\mathrm{Fi}'_{24}, \mathrm{Fi}_{22}, \mathrm{O'N}, HN$, where they are the retained extensions of index two. The ordinary tables and their actual class fusions verify the condition in Lemma 2 in each of those four cases. For the rows with $R = 0$, the inverse class differs from the chosen class in the full automorphism group, not merely in an unspecified subgroup.

For clarity, the nonzero values of μ actually used are

G	μ	G	μ
J_4	3499519	Fi_{22}	1436095
Ly	406875	$\mathrm{O'N}$	3095
Fi'_{24}	25790392063	HN	140151
Ru	13455		

These are maxima over complete maximal-subgroup lists, as specified in (3) or (4). For example, the largest value for the automorphism group of Fi'_{24}, after omitting its simple socle, is $i_2(2 \times \mathrm{Fi}_{23}) = 25790392063$. Thus even this coarse bound suffices. No assertion that an inverter belongs to a particular class of involutions is needed.

Theorem 1 gives the required generating pairs. For the polyhedron conclusion put $y = x^{-1}t$. The cyclic intersection $\langle x \rangle \cap \langle y \rangle$ is central in $\langle x, y \rangle = G$, so it is trivial. Also $(xy)^2 = 1$. The element y cannot be an involution: otherwise G would be generated by the two involutions y, t and would be dihedral. The case $y = 1$ would make G cyclic. Hence $|x|, |y| > 2$. The retained Schulte–Weiss reconstruction theorem supplies a chiral polyhedron with full automorphism group G, since the reversing automorphism has been excluded.

3 Inputs, verification, and attribution

The source's existing existence conclusions are being reproved, not claimed as new. The retained facts are the identifications of the sporadic groups and their automorphism groups, the complete lists of maximal subgroup classes, the ordinary class sizes and element orders, and the displayed inclusion fusions. The verifier obtains the numerical data from GAP's Character Table Library, in its recorded edition. Its `Maxes` attribute is documented to give one representative for every conjugacy class of maximal subgroups when it is present. Completeness of those classification lists is a premise; it is not proved by iterating over a software list.

For every required table, the checker sums the sizes of the classes of exact order two or p. It retains every maximal-subgroup class, including repetitions of a table name when distinct subgroup classes have the same character table. It verifies each centralizer index used in (4), the actual inverse-class test in (2), and all fourteen strict inequalities with exact integers or rationals. The full underlying rows and their source bindings are retained, so the small displayed table is not an unexplained set of trusted totals.

Together with the separately reviewed twelve explicit word certificates, these fourteen cases cover all twenty-six sporadic groups in the source theorem. The alternating branch has its own uniform proof. The classical and exceptional Lie families remain separate obligations.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, fixed arXiv:1603.07713v3, Sections 2 and 6.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991),493–516.

[3] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[4] T. Breuer, *The GAP Character Table Library*, version 1.3.11, data and manual Section 3.7.1; the exact retained installation files are specified in the dependency ledger.